

DEEPAKRAJ B

FORWARD DEPLOYED ENGINEER | AI INTEGRATIONS, BACKEND (PYTHON / JAVA) & CLOUD

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📄 LinkedIn

PROFILE

Forward-deployed Integration and Backend Software Engineer with hands-on experience delivering production client integrations, on-site defense deployments, AI/LLM orchestration pipelines, and scalable APIs across Python and Java. Proven track record deploying on-site at client facilities under strict operational constraints to autonomously build and deliver end-to-end full-stack and IoT systems. Skilled in **RESTful APIs, WebSockets, webhooks**, and integrating LLM workflows using **LangChain, LangGraph, and OpenAI SDKs** with cloud infrastructure on **AWS & GCP**. Experienced in Docker containerization, **OAuth2 / JWT** security, and client-facing system delivery.

PROFESSIONAL EXPERIENCE

Software Engineer | Kritilabs Technologies Pvt. Ltd., Chennai, India July 2024 - Present

Sentinel Kote - Offline RFID Weapon Tracking System (Indian Army)

Java, Android, React, RFID Hardware, Offline Systems

- Forward-deployed on-site at an **Indian Army camp**, gathering operational requirements directly from soldiers to replace error-prone manual ledger workflows.
- Architected and delivered the complete **offline RFID weapon inventory platform** (Android app, React UI, and Java backend) solo within 2 weeks.
- Integrated hardware RFID scanners with offline-first synchronization and implemented guided UI walkthroughs for rapid user onboarding in field conditions.

AI Integration & Natural Language Query Platform

Python, FastAPI, LangGraph, LangChain, OpenAI APIs, Docker, AWS

- Built and deployed an enterprise **NL2SQL integration engine** using **Python, LangGraph, and FastAPI**, enabling natural-language querying across multi-table SQL databases.
- Implemented tool-calling orchestration with schema metadata retrieval (**ChromaDB**), multi-step validation, and prompt design to ensure high query execution accuracy.
- Containerized Python services via **Docker** and established automated **CI/CD deployment pipelines** on **AWS (EC2, S3)** for repeatable, zero-downtime releases.

Enterprise Client Systems Integration & Platform Rollout

Python, Frappe, REST APIs, Webhooks, MySQL, Postman

- Led platform integration across 4 manufacturing units (700+ users), syncing production planning, inventory, and QA systems.
- Developed **REST API integrations and webhook receivers** connecting legacy ERP, inventory databases, and Android barcode scanners with Frappe backend APIs, cutting manual data entry by **80%**.
- Optimized **MySQL & PostgreSQL** database schemas, query indexes, and API endpoints, improving throughput by **30%**.

Real-Time Client Telemetry & Microservices Integration

Java, Spring Boot, WebSockets, REST APIs, Redis, JWT

- Engineered real-time telemetry ingestion pipelines and microservices for live vehicle fleet tracking and smart lock monitoring for **IOCL and BPCL**.
- Implemented secure authentication with **JWT and OAuth2** dynamic RBAC, and deployed backend analytics for **Indian Railways**.

TECHNICAL SKILLS

Programming Languages: Python (FastAPI, Django, Flask), Java (Spring Boot), SQL, JavaScript

AI & LLM Orchestration: LangChain, LangGraph, OpenAI APIs, Prompt Design, NL2SQL, ChromaDB

Databases & Caching: PostgreSQL, MySQL, Redis, SQLAlchemy, Query Optimization

Integrations & Protocols: RESTful APIs, WebSockets, Webhooks, OAuth2, JWT, IoT Gateways

Cloud & DevOps: AWS (EC2, S3), GCP, Docker, CI/CD Pipelines, Linux, Nginx

Testing & Observability: Postman, Pytest, JUnit 5, Swagger/OpenAPI, Git, JIRA

EDUCATION

B.Tech in Computer Science & Engineering | SASTRA Deemed University, Thanjavur, Tamil Nadu **CGPA: 7.2955** | 2020 - 2024

CERTIFICATIONS

- Python Crash Course - Google
- Machine Learning Specialization
- Software Engineering Specialization - IBM
- Backend Development Specialization - Meta